

axis), forming a 3D acceleration vector. This 3D acceleration vector points to the orientation and direction of gravity. It can be used for many applications. For example, if you are tossing a phone up in the air, then all the values (acceleration along the X, Y and Z axes) will fall to 0, so you could design an application that makes the phone scream if you drop it.

One of the most useful applications based on free fall detection is the protection of hard disk drives (HDD). In case the mobile/PDA, laptop or notebook is dropped, the HDD head can slide over the drive, erasing the data. Using MEMS motion sensing, a safety signal can be generated to park away the HDD head and save all the data.

Accelerometers are featured in Lenovo's 'active protection system' and also in Apple's 'sudden motion sensor,' which detects a drop. In a paper presented in Planet Analog by Bill Raasch, VP, market development for computers and peripherals, STMicroelectronics, he says, "By anticipating a free-fall incident, you can design into your systems a 'safe' condition that will improve the likelihood of saving precious data. Once the accelerometer is integrated into a portable device, it can enable other possible applications that may need to know which way is 'down,'"

Shock detection. Shock is a sudden change in acceleration or instantaneous acceleration. Shock detection is used in vehicle safety systems for crash detection or deployment of airbags. Shock or impact sensors are very carefully chosen as they demand high sensitivity and quick response times. Programmable shock sensors can be used in detecting sudden irregularities in the heart rate and blood pressure. It also helps in early diagnosis by continuous monitoring.

Vibration detection. Highly sensitive vibration sensors can be used to detect slight vibrations in vehicle misalignment, worn-out bearings, a mismatch in other machinery components, etc. The control action/signal generated from the main processor can help in issues like this. MEMS-based vibration sensors

Useful theory behind MEMS motion sensors

Accelerometers are microscopic silicon chips that can detect motion. They consist of a proof-mass (weighs a fraction of a gram) that experiences inertial force (D'Alembert's force). The force causes the proof-mass to move, thereby elastically deforming the suspension. An accelerometer can measure acceleration in three directions (the X, Y and Z axes) and the gyroscope can measure rotation along three axes (yaw, pitch and roll). The sensing mechanism decides the type of accelerometer, viz., piezo-resistive, capacitive, piezo-electric, tunnelling, optical, resonant, thermal, etc.

Signal conditioning is used for converting the signal to an amplified analogue or digital signal. Thus the mechanical part used in sensing and the signal processing part forms the electronics of an MEMS chip. MEMS can be classified depending on the type of output—analogue or digital. Normally, in an application/product, MEMS accelerometers/gyroscopes are used in combination with microcontrollers. Analogue output type MEMS can be used with microcontrollers with an in-built ADC, or the digital version can be used. It's easier to use digital MEMS as users do not have to worry about data conversion.

MEMS sensor generations

1st generation. These were based on a silicon structure. It was a multi-chip approach where the sensing element was one chip and the signal processing electronics was on other chip. It gave an analogue output with some amplification on the microchip.

2nd generation. These combined the analogue amplification with an analogue-to-digital converter on a microchip.

3rd generation. In this phase, digital intelligence for linearisation and temperature compensation were added.

4th generation. These included programmable MEMS, memory cells for calibration and temperature compensation.

are used in vehicles, along with other motion sensing, in tire pressure measurement (TPM) and electronic stability control systems (ESC). These are also used for fine tuning various equipment. Vibrations are typically measured using analogue sensing elements.

Change of orientation. The pioneer in the use of 'change of orientation' features was Apple's iPhone which enabled changing from portrait to landscape. Change of orientation is a concept that can be very creatively used in many high-end hand-held electronic devices. Since the X, Y and Z values represent acceleration in different directions, when the phone is working in a plane, one of the values remain constant. It is possible to detect the face up/down of a phone and this information could be used for power saving applications, changing profiles when in a meeting and many other functions. Change of orientation is very conveniently used in cameras for art photography when creating illusions and different perceptions. The screen would change the arrangement

of the movable optics, giving a different angle to the image frame.

Image stabilisation. The sleek and sophisticated new-generation digital cameras that are thinner and offer better performance are highly susceptible to image blurring due to natural hand shaking. Hand jitter results in rotational frequencies of 1 to 20Hz. Optical image stabilisation is obtained by detecting these minute vibrations using gyroscopes and taking corrective actions. Most of the latest cameras provide better image stability features by using MEMS motion sensing.

Accelerometers are also used in the new LCD projectors and monitors for keystone correction. The keystone effect occurs when the projected image is distorted as a trapezoid. This is caused when the projection is tilted. The angle of tilt is measured and corrective action is taken by the processor to adjust the lens and correct the projected image.

Typical application areas

To select a MEMS sensor that best fits in an application, a designer has to